FINANCIAL FRAUD DETECTION

PROJECT REPORT

Data Science & Analytics Internship

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Zidio Development

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Project Context: Financial Fraud Detection — Zidio Development

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1. Project Overview

Purpose

The Financial Fraud Detection project is an end-to-end machine-learning and analytics workflow for identifying suspicious transactions, estimating fraud probability, classifying risk and presenting transaction-level monitoring information. The implemented workflow combines data preparation, exploratory analysis, feature engineering, supervised classification, risk mapping and Power BI investigation views.

Business Context

Financial transaction datasets can contain patterns that are difficult to identify consistently through manual review. The project uses transaction amount, customer behaviour, payment method, device, location, international status and time-related variables to create a structured fraud-screening workflow.

Project Outcomes

- Prepared and analysed 5,000 transaction records, including 482 labelled fraudulent transactions.
- Measured a fraud rate of 9.64% across the supplied dataset.
- Created a 29-column feature-engineered dataset for supervised classification.
- Compared Logistic Regression, Decision Tree, Random Forest and Gradient Boosting.
- Generated fraud probabilities and mapped monitored transactions into Low Risk, Medium Risk and High Risk levels.
- Produced monitoring outputs containing state, action and alert status.
- Developed a multi-page dashboard covering transaction, merchant, location, customer, time, probability, risk, model, reinforcement-learning, monitoring and investigation views.

Project Scope

The implemented scope is based on the five supplied Python/Jupyter notebooks, the generated CSV outputs and the 12-page Power BI dashboard. The workflow covers stored-data preparation, analysis, supervised fraud classification, probability-based risk categorisation, monitoring simulation and transaction investigation.

2. Business Problem & Objectives

Business Problem

- Financial institutions need to identify suspicious transactions quickly so that potentially fraudulent activity can be reviewed.
- Large transaction volumes make manual inspection of every record inefficient.
- Fraudulent behaviour may depend on several variables at the same time, including transaction amount, customer spending behaviour, payment method, location and device.
- A classification model can provide a consistent predicted fraud status and probability for downstream risk monitoring.
- A dashboard is required to convert model output into understandable risk, alert and investigation views.

Objectives

- Clean and prepare the supplied financial transaction data.
- Perform exploratory analysis of transaction amount, merchant category, payment method, location, customer behaviour and time.
- Engineer numerical, temporal and categorical features suitable for machine-learning models.
- Train and compare multiple supervised classification algorithms.
- Evaluate models using Accuracy, Precision, Recall, F1-Score and ROC-AUC.
- Generate fraud probability and risk-level outputs for monitored transactions.
- Create alert statuses and review/block actions from risk outputs.
- Present the results through interactive dashboard pages for monitoring and investigation.

Functional Scope

Area	Function
Data	Cleaning, validation, transformation and feature preparation
Machine Learning	Binary fraud classification and probability estimation
Risk	Low / Medium / High risk categorisation
Monitoring	Transaction-level state, action and alert status
Dashboard	Fraud analysis, model performance, monitoring and investigation
Monitoring implementation	Stored CSV monitoring simulation and refreshable dashboard output

3. Data & Dataset Summary

Dataset Characteristics

The supplied financial dataset contains transaction-level records with customer, transaction, merchant, payment, device, location, behavioural and fraud-label fields. The cleaned dataset is the base for feature engineering and model development.

Dataset	Rows	Columns	Primary purpose
financial_fraud_clean	5,000	14	Clean transaction-level analytical data
fraud_features	5,000	29	Engineered machine-learning features
fraud_predictions	1,000	32	Prediction, probability and risk outputs
fraud_monitoring	1,000	7	Monitoring, RL action and alert outputs
model_comparison	4	6	Model evaluation metrics

Key Transaction Statistics

Measure	Value
Total transactions	5,000
Fraudulent transactions	482
Non-fraudulent transactions	4,518
Fraud rate	9.64%
Average transaction amount	79.18
Average fraudulent transaction amount	81.99
Average non-fraudulent transaction amount	78.88
International transactions	441 (8.82%)

The report uses the implemented CSV outputs as the evidence base. The analytical workflow is stored-data based and is executed through the supplied notebooks rather than through a live transaction-streaming service.

4. Data Preparation & Exploratory Analysis

Data Preparation

- Transaction date values were converted to datetime values for time-based analysis.
- The raw CSV was loaded with Pandas and checked for its 5,000 rows, 14 columns and required transaction, customer, behavioural and fraud-label fields.
- Categorical variables were converted into model-ready indicator variables during feature engineering.
- Amount_vs_Average was created to compare each transaction amount with the customer average spend.
- The cleaned transaction data was exported as financial_fraud_clean.csv and used as the input for the subsequent EDA and feature-engineering notebooks.

Exploratory Findings

The dashboard evidence shows 5K total transactions and 482 fraudulent transactions. Fraud patterns were examined across merchant category, payment method, location, customer behaviour and time.

Observation	Result
Fraudulent transactions	482
Fraud rate	9.64%
Highest fraud count by merchant category	Food — 74
Next merchant category	Entertainment — 72
Highest fraud count by payment method	Debit Card — 103
Highest fraud count by location	Mumbai — 78
Suspicious Keyword = Yes	260 records; 123 fraudulent

Interpretation

These counts describe the supplied dataset and should not be interpreted as causal effects. Category, payment and location distributions are useful for understanding transaction patterns and for prioritising further investigation.

5. Feature Engineering & Model Inputs

Feature Engineering

The feature-engineered dataset contains 29 columns. It combines continuous transaction and customer variables with temporal indicators and one-hot encoded categorical variables. This representation converts the original transaction fields into numerical inputs suitable for supervised classification.

Feature group	Included variables
Numerical / behavioural	Transaction_Amount, Is_International, Previous_Transactions, Average_Spend, Account_Age_Days
Temporal	Transaction_Hour, Transaction_Month
Behavioural ratio	Amount_vs_Average
Merchant encoding	Entertainment, Fashion, Food, Grocery, Health, Travel, Utilities
Payment encoding	Debit Card, NetBanking, PayPal, UPI
Device encoding	Mobile, POS
Location encoding	Chennai, Delhi, Hyderabad, Kolkata, Mumbai, Pune
Target	Fraudulent

Why These Features Matter

- Transaction_Amount represents the monetary size of the transaction.
- Previous_Transactions and Average_Spend provide customer-behaviour context.
- Account_Age_Days captures account-history information.
- Transaction_Hour and Transaction_Month provide time context.
- Amount_vs_Average captures how unusual the transaction amount is relative to average spending.
- Merchant, payment, device and location indicators allow the classifiers to distinguish categorical transaction patterns.

6. Fraud Detection Methodology

Classification Approach

Four supervised classification models were compared on the supplied fraud-classification task. The objective was to predict the Fraudulent label and produce a probability output that could be translated into operational risk levels.

Model	Role in project
Logistic Regression	Linear baseline for binary fraud classification
Decision Tree	Tree classifier capturing nonlinear decision splits
Random Forest	Ensemble of multiple decision trees
Gradient Boosting	Sequential boosting ensemble for classification

Fraud Probability & Risk

The prediction dataset contains Predicted_Fraudulent, Fraud_Probability and Risk_Level. The monitoring dataset then maps the probability/risk output into a monitoring state, action and alert status.

Risk level	Monitoring action	Alert status	Count
Low Risk	Approve	Normal	914
Medium Risk	Review	REVIEW	82
High Risk	Block	ALERT	4

Monitoring Logic

- Low-risk transactions are represented with an approval action and Normal alert status.
- Medium-risk transactions are routed to Review and receive a REVIEW alert.
- High-risk transactions are routed to Block and receive an ALERT status.
- The supplied monitoring file contains 1,000 monitored transaction records.

7. Model Evaluation

Evaluation Method

The supplied model_comparison file evaluates four supervised classifiers using Accuracy, Precision, Recall, F1-Score and ROC-AUC. These metrics provide a broader view than Accuracy alone, which is important because the dataset contains substantially more non-fraudulent than fraudulent transactions.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	90.30%	48.72%	19.79%	28.15%	87.63%
Decision Tree	85.60%	28.95%	34.38%	31.43%	62.71%
Random Forest	90.50%	52.63%	10.42%	17.39%	89.67%
Gradient Boosting	90.30%	48.39%	15.62%	23.62%	90.01%

Metric Interpretation

- Accuracy measures the overall proportion of correct classifications.
- Precision measures how many transactions predicted as fraud were actually fraudulent.
- Recall measures how many actual fraudulent transactions were detected.
- F1-Score balances Precision and Recall.
- ROC-AUC measures discrimination across classification thresholds.

In the supplied comparison, Random Forest records the highest Accuracy (90.50%) and Precision (52.63%). Decision Tree records the highest Recall (34.38%) and F1-Score (31.43%). Gradient Boosting records the highest ROC-AUC (90.01%). These results illustrate different metric trade-offs rather than a single universal performance measure.

8. Implementation Workflow & Technology Stack

Area	Technology / approach	Purpose
Programming	Python	Data analysis and machine-learning workflow
Data handling	Pandas, NumPy	Cleaning, transformation and tabular processing
Visualization	Matplotlib, Seaborn	EDA and model-evaluation visualizations
Machine Learning	Scikit-learn	Scaling, classification and evaluation metrics
Data Storage / ETL	CSV / Pandas	Cleaned transaction files and intermediate/output data exchange
Monitoring	Pandas / saved CSV outputs	Stored-data risk, state, action and alert simulation
Dashboard	Power BI	Interactive visual analysis and monitoring
Notebook workflow	Jupyter Notebook	Step-by-step project implementation
Versioning	Git / GitHub	Project version control

Implemented workflow: raw CSV → cleaning → EDA → feature engineering → model training/evaluation → fraud probability → risk levels → monitoring simulation → Power BI dashboard.

Project Structure

- data/ — cleaned transaction data, engineered features, predictions and monitoring outputs.
- notebooks/ — data preparation, feature engineering, model training and evaluation workflow.
- dashboard/ — Power BI dashboard pages and visual analysis.
- README.md — project overview and usage information.

Implementation Note

The implemented workflow is CSV- and notebook-based. The five notebooks form a sequential pipeline from raw data preparation to EDA, feature engineering, model training/evaluation, prediction, risk categorisation and monitoring analysis. The separate TXT specification contains example SQLite and Kafka/Spark code, but those examples are not used as implementation evidence for this project.

9. Dashboard Overview

Dashboard Purpose

The Power BI dashboard is organized into twelve focused pages. Together they move from overall fraud volume and transaction patterns to merchant, location, customer and time analysis, then to probability, risk, model performance, reinforcement learning, stored-data monitoring and fraud investigation.

What We Did in Power BI

- Created a consistent professional visual identity across the financial fraud report.
- Added KPI cards for transaction volume, fraudulent transactions, fraud rate and high-risk predictions.
- Connected charts for risk distribution, fraud by category, transaction analysis and monitoring.
- Added model-performance views comparing Accuracy, Precision, Recall, F1-Score and ROC-AUC.
- Added monitoring and reinforcement-learning views that connect risk outputs with actions and alerts in the saved analytical outputs.
- Added a transaction-level fraud investigation page for detailed review.

Dashboard Pages

Page	Primary purpose
Fraud Detection Overview	Overall fraud volume, fraud rate, risk distribution and category view
Transaction Analysis	Transaction amount distribution and monthly activity
Merchant & Payment Analysis	Fraud by merchant category and payment method
Location Analysis	Fraud distribution by location and international activity
Customer Behavior	Customer spend, history and account-age patterns
Time Analysis	Hourly, day-part and monthly transaction patterns
Fraud Probability Analysis	Probability distribution and predicted/actual fraud
Risk Analysis	Risk-level distribution and fraud by risk
ML Model Performance	Model metric comparison
Reinforcement Learning	State, action and alert distribution
Real-Time Monitoring	Transaction monitoring and alert status
Fraud Investigation	Detailed transaction-level investigation

10. Fraud Detection Overview Dashboard

Provides the management-level starting point for reviewing transaction volume, fraud incidence and risk distribution.

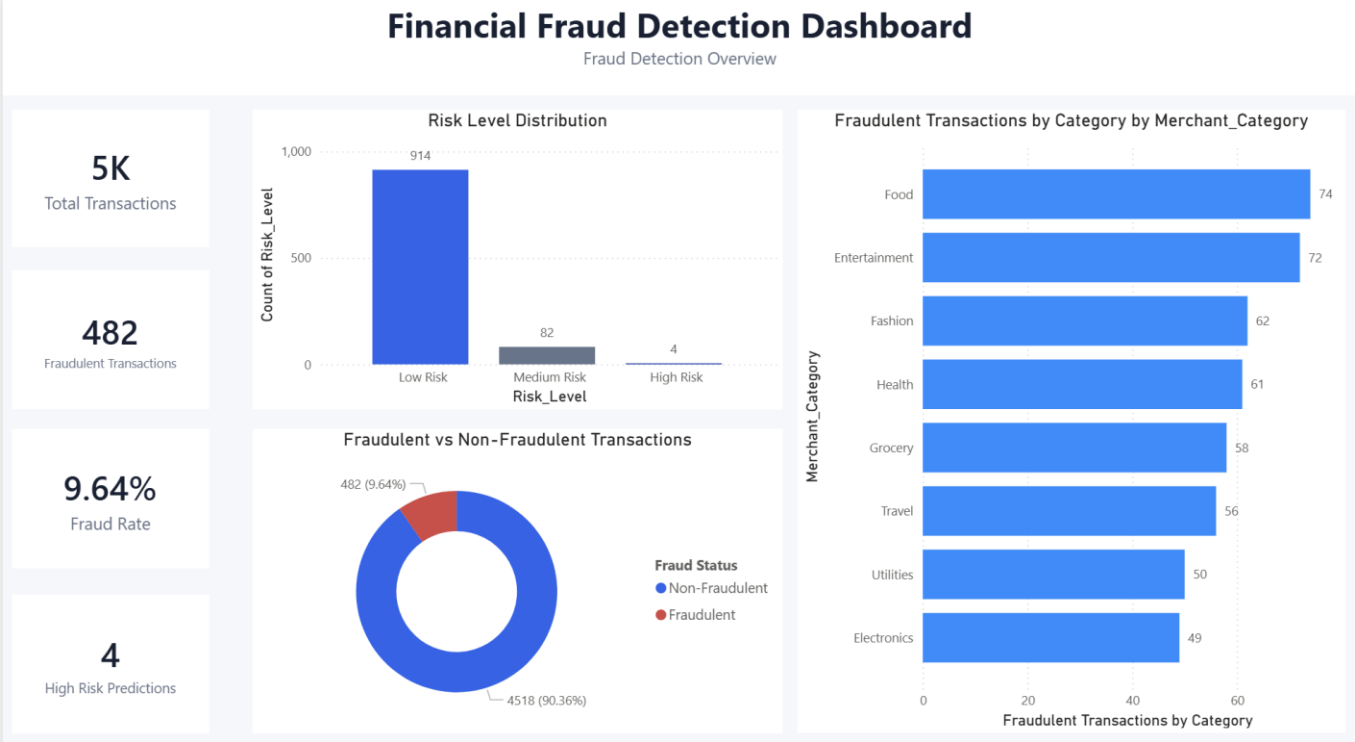


Figure 1. Fraud Detection Overview Dashboard

Key information

- 5K total transactions and 482 fraudulent transactions
- 9.64% fraud rate shown in the overview
- Low, Medium and High risk distribution
- Fraudulent transaction counts by merchant category

Business use

Supports a quick review of overall fraud exposure and directs attention toward risk categories and merchant segments requiring further investigation.

11. Transaction Analysis Dashboard

Focuses on transaction amount behaviour and overall transaction volume over time.

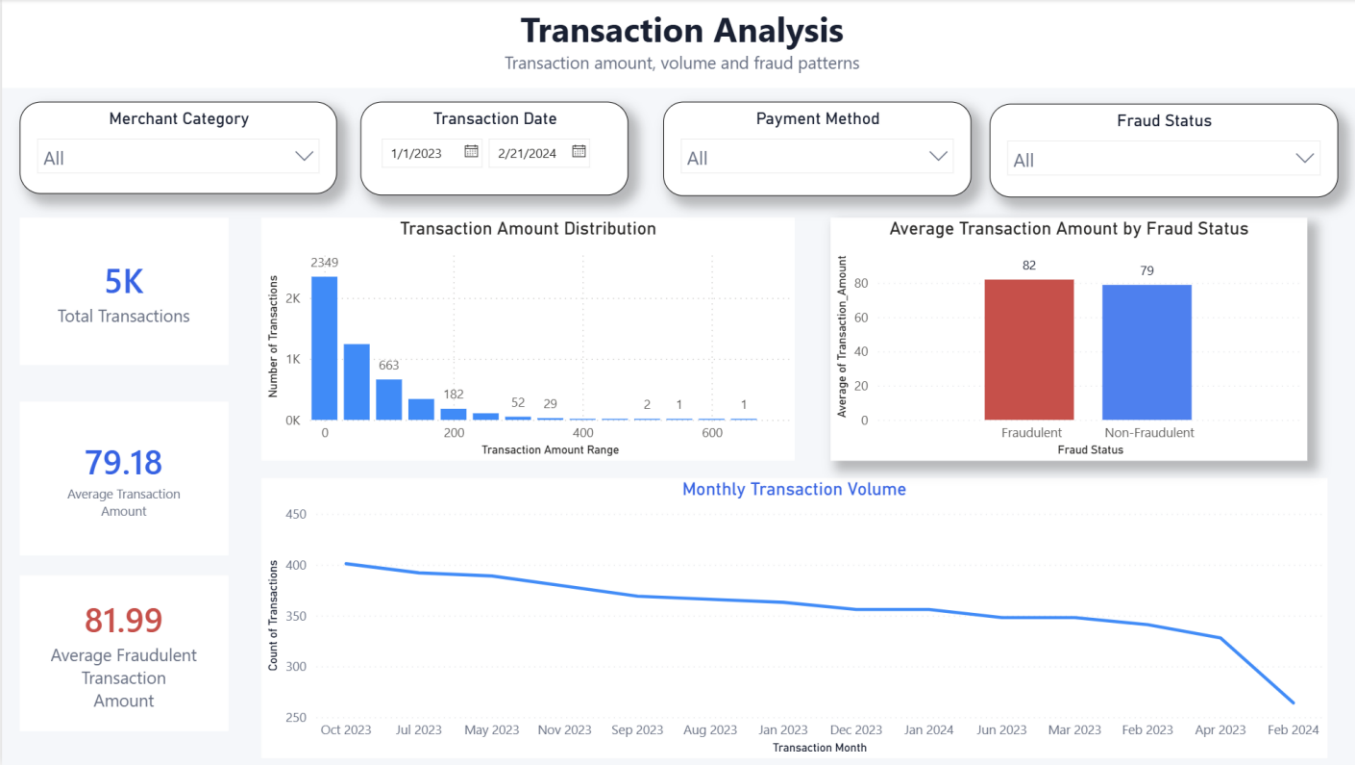


Figure 2. Transaction Analysis Dashboard

Key information

- Average transaction amount of 79.18
- Average fraudulent transaction amount of 81.99
- Transaction amount distribution
- Monthly transaction-volume trend

Business use

Supports exploration of transaction-value patterns before moving to more detailed fraud, risk or investigation views.

12. Merchant & Payment Analysis Dashboard

Examines fraudulent activity across merchant categories and payment methods.

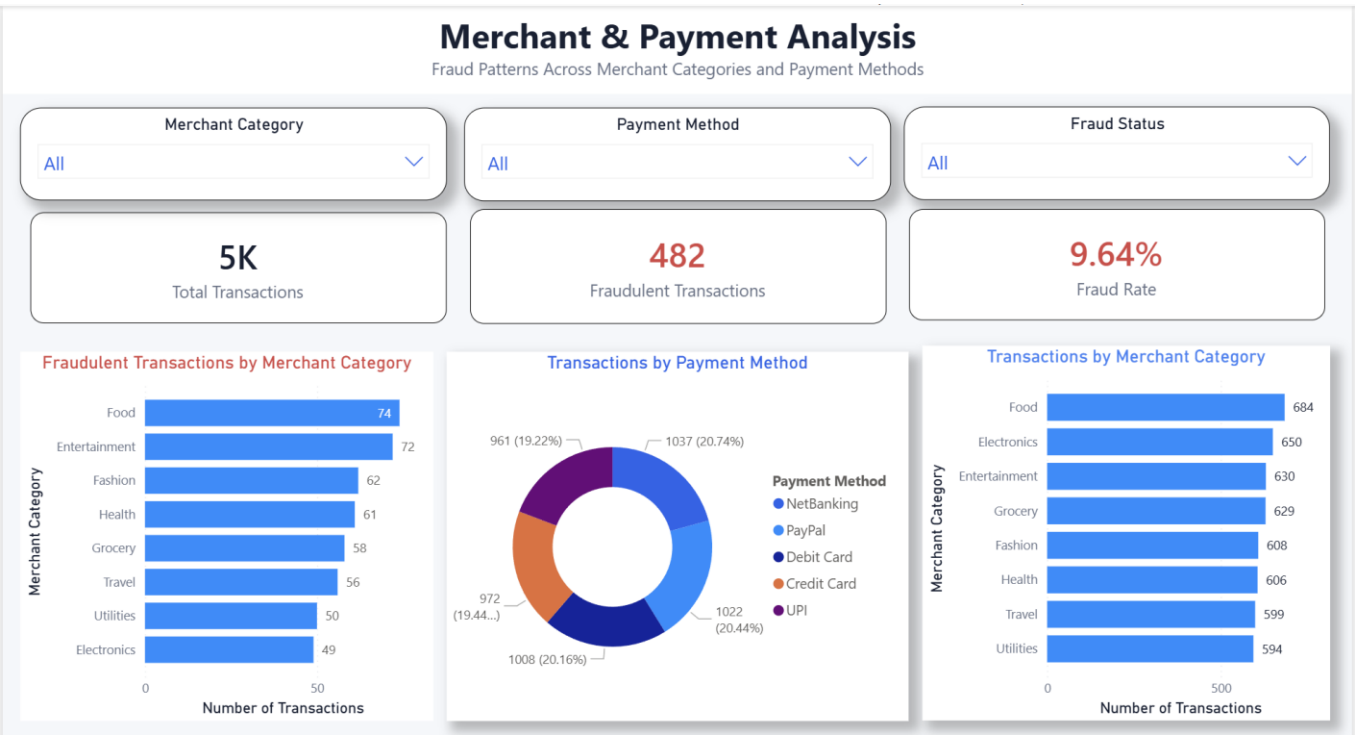


Figure 3. Merchant & Payment Analysis Dashboard

Key information

- Fraud counts by merchant category
- Fraud distribution across payment methods
- Merchant-category transaction volume
- Comparison of fraud and non-fraud activity

Business use

Helps analysts identify merchant and payment segments where transaction patterns warrant additional review.

13. Location Analysis Dashboard

Presents the geographic distribution of transactions and fraudulent activity, including international status.

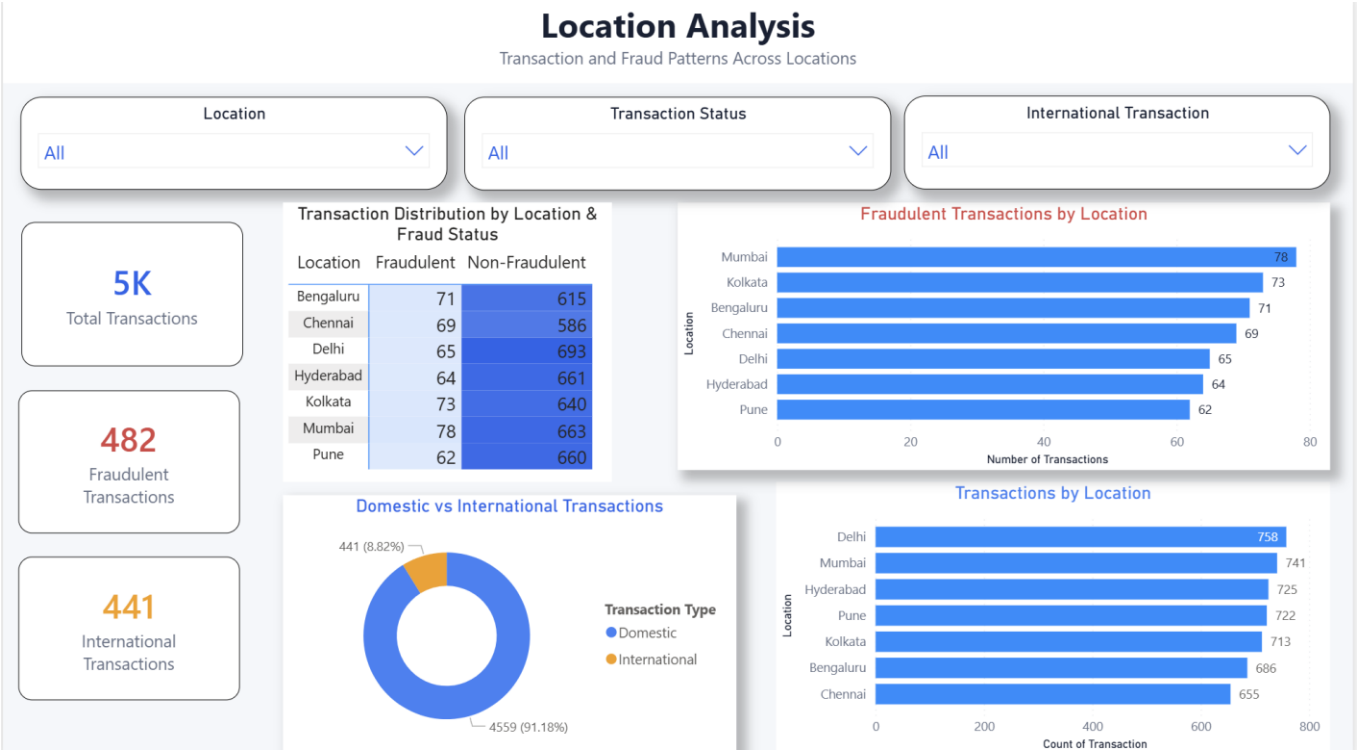


Figure 4. Location Analysis Dashboard

Key information

- Fraud distribution by location
- Transaction volume by location
- Domestic versus international activity
- Highest fraud count by location: Mumbai — 78

Business use

Supports geographic investigation by highlighting locations with larger transaction or fraud volumes.

14. Customer Behavior Dashboard

Connects customer spending behaviour, previous transaction history and account age with fraud status.

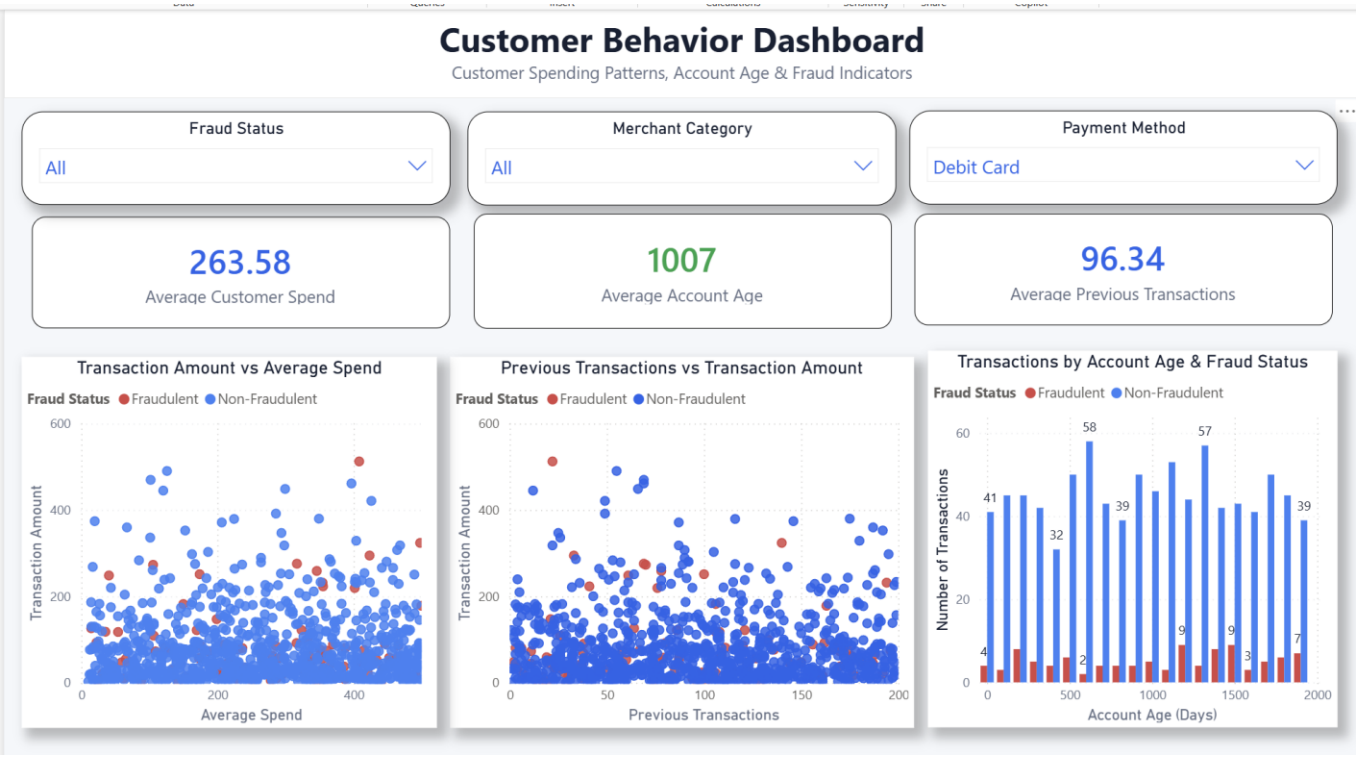


Figure 5. Customer Behavior Dashboard

Key information

- Average customer spend patterns
- Previous-transaction behaviour
- Account-age distribution
- Transaction amount relationships by fraud status

Business use

Provides behavioural context for fraud analysis and supports investigation of unusual customer activity.

15. Time Analysis Dashboard

Examines transaction activity across hours, day-parts and months.



Figure 6. Time Analysis Dashboard

Key information

- Transaction volume by hour
- Day-part comparison
- Monthly transaction trends
- Hourly fraud versus non-fraud activity

Business use

Supports temporal analysis and helps analysts identify periods where transaction behaviour changes.

16. Fraud Probability Analysis Dashboard

Explores model-generated fraud probabilities and the relationship between predicted and actual fraud.

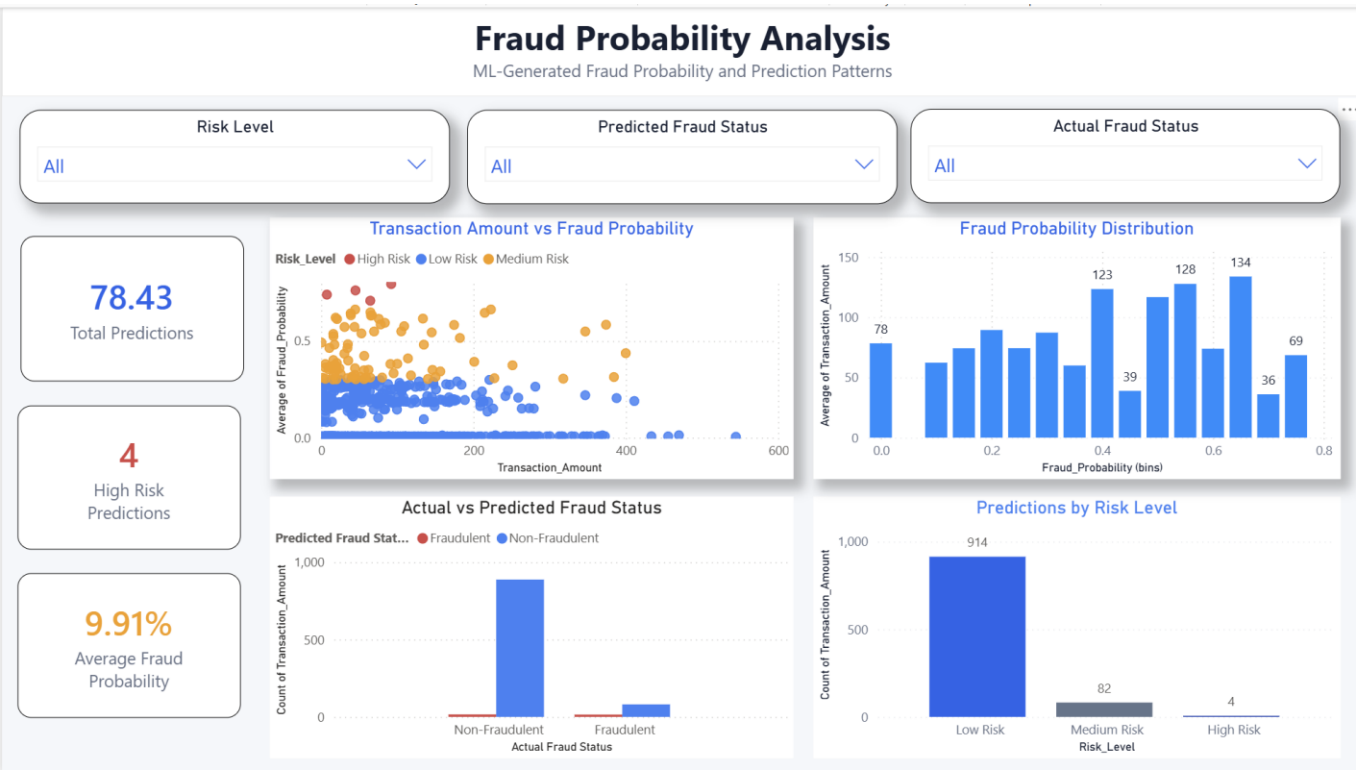


Figure 7. Fraud Probability Analysis Dashboard

Key information

- Fraud probability distribution
- Transaction amount versus fraud probability
- Predicted versus actual fraud
- Predictions grouped by risk

Business use

Connects the classifier output to interpretable probability and risk views for analysts.

17. Risk Analysis Dashboard

Summarizes Low, Medium and High risk transactions and compares fraud outcomes by risk category.

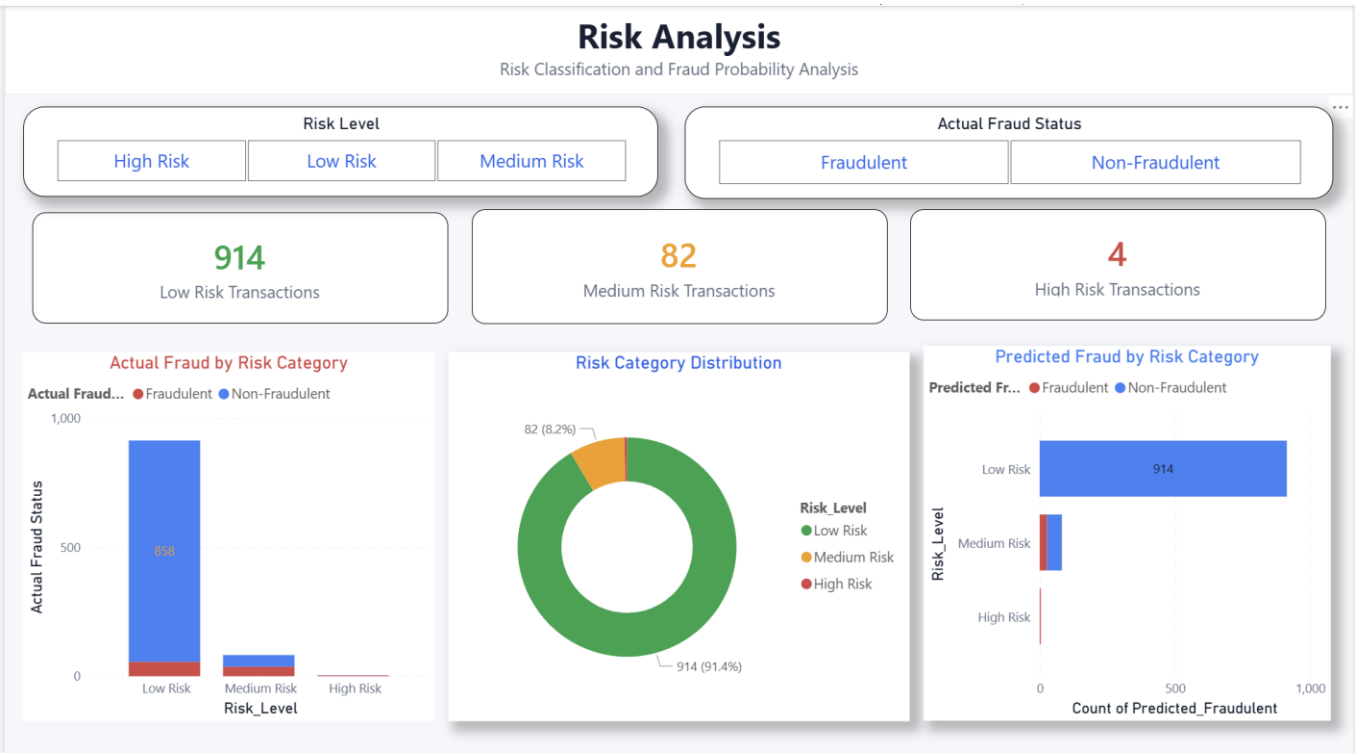


Figure 8. Risk Analysis Dashboard

Key information

- Risk-level distribution
- Actual fraud by risk category
- Predicted fraud by risk
- Risk categories linked to monitoring actions

Business use

Provides an operational view of how model outputs translate into risk segments and review priorities.

18. ML Model Performance Dashboard

Compares the four classification models across the evaluation metrics supplied with the project.

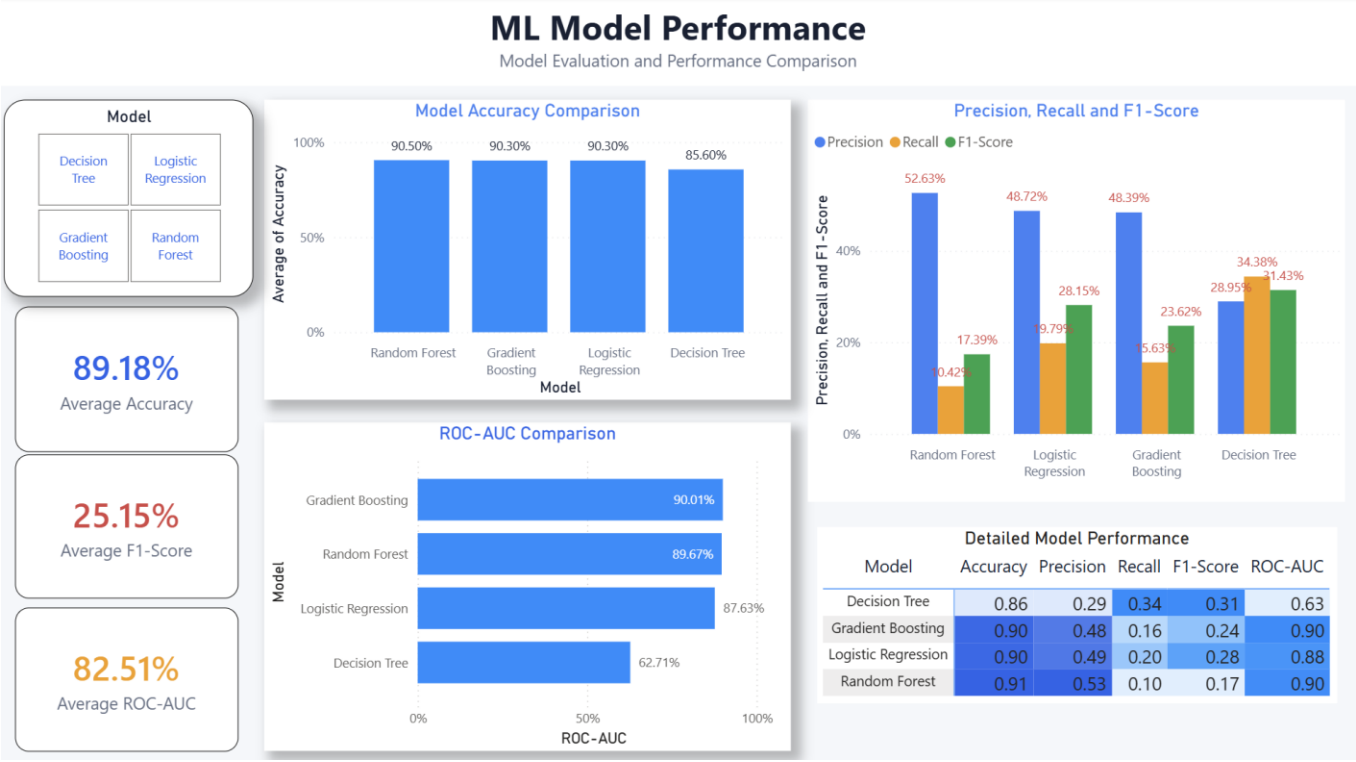


Figure 9. ML Model Performance Dashboard

Key information

- Accuracy comparison
- Precision, Recall and F1-Score comparison
- ROC-AUC comparison
- Four-model performance view

Business use

Supports transparent model evaluation by showing that different algorithms have different metric profiles.

19. Reinforcement Learning Dashboard

Visualizes the reinforcement-learning state and action outputs contained in the monitoring dataset.



Figure 10. Reinforcement Learning Dashboard

Key information

- RL state distribution
- RL action distribution
- Actions by state
- Alert-status distribution

Business use

Shows how monitored transactions are mapped into approve, review and block actions in the supplied monitoring output.

20. Real-Time Monitoring Dashboard

Provides a monitoring-oriented view of transaction counts, risk levels, actions and alert statuses.

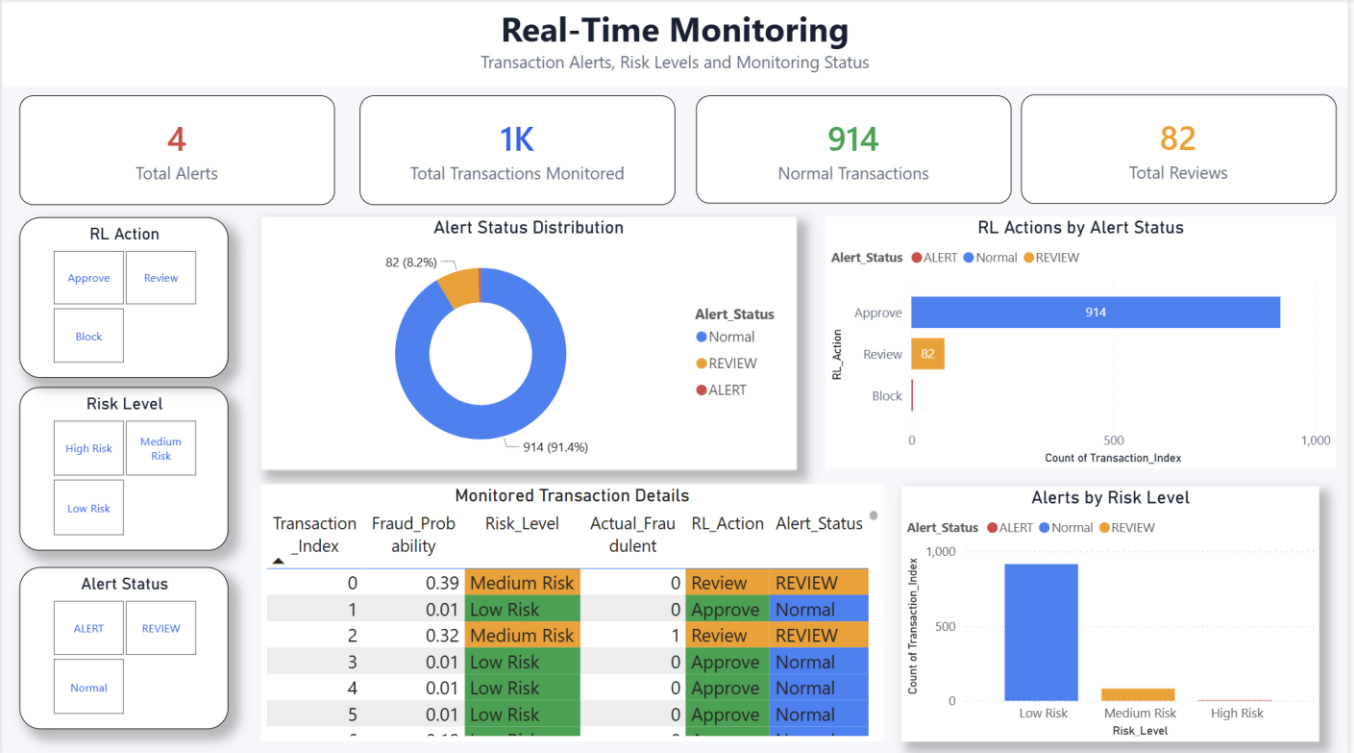


Figure 11. Real-Time Monitoring Dashboard

Key information

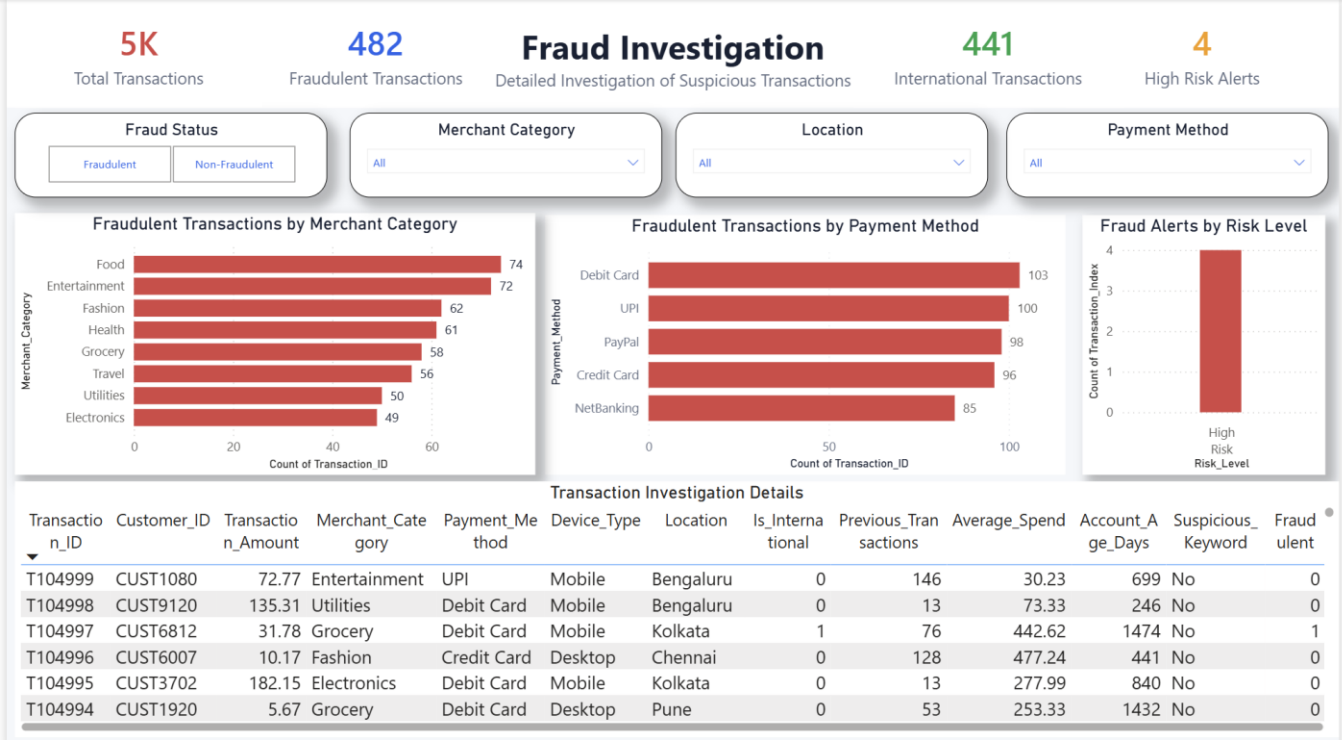
- 1,000 monitored transactions
- 914 Normal transactions
- 82 REVIEW cases
- 4 ALERT cases

Business use

Supports transaction monitoring by connecting risk levels with operational actions and alert status.

21. Fraud Investigation Dashboard

Provides a detailed investigation view of suspicious and fraudulent transaction records.



Transaction_ID	Customer_ID	Transaction_Amount	Merchant_Category	Payment_Method	Device_Type	Location	Is_International	Previous_Transactions	Average_Spend	Account_Age_Days	Suspicious_Keyword	Fraudulent
T104999	CUST1080	72.77	Entertainment	UPI	Mobile	Bengaluru	0	146	30.23	699	No	0
T104998	CUST9120	135.31	Utilities	Debit Card	Mobile	Bengaluru	0	13	73.33	246	No	0
T104997	CUST6812	31.78	Grocery	Debit Card	Mobile	Kolkata	1	76	442.62	1474	No	1
T104996	CUST6007	10.17	Fashion	Credit Card	Desktop	Chennai	0	128	477.24	441	No	0
T104995	CUST3702	182.15	Electronics	Debit Card	Mobile	Kolkata	0	13	277.99	840	No	0
T104994	CUST1920	5.67	Grocery	Debit Card	Desktop	Pune	0	53	253.33	1432	No	0

Figure 12. Fraud Investigation Dashboard

Key information

- Fraudulent transaction analysis by merchant category
- Payment-method and risk-level views
- Transaction-level detail table
- Customer, amount, merchant, payment, device and location fields

Business use

Supports analyst investigation by moving from aggregate fraud patterns to individual transaction records.

22. Challenges, Learnings, Limitations & Future Improvements

Challenges & How They Were Handled

- Class imbalance is visible in the dataset: 482 fraudulent records versus 4,518 non-fraudulent records. This makes Recall, Precision and F1 important alongside Accuracy.
- Different classifiers show different metric profiles, so model comparison was performed across five metrics instead of using Accuracy alone.
- Feature engineering was required to convert categorical variables and behavioural information into model-ready numerical inputs.
- The dashboard had to connect model outputs with operational concepts such as risk level, review and alert status.
- The project specification includes streaming and alerting components that require additional infrastructure beyond the supplied CSV-based analytical outputs.

Key Learnings

- Built a complete workflow from cleaned transaction data to machine-learning predictions and dashboard analysis.
- Used feature engineering to represent transaction, customer, temporal and categorical patterns.
- Compared four supervised classification algorithms using multiple evaluation metrics.
- Converted fraud probability into practical risk categories and monitoring actions.
- Used dashboard views to move from aggregate fraud patterns to transaction-level investigation.

Limitations

- The supplied dataset contains 5,000 transactions, so conclusions are limited to this dataset.
- The fraud label is treated as the target; the report does not establish causal relationships between individual features and fraud.
- The monitoring output contains 1,000 records and should be interpreted as the supplied monitoring sample rather than a production transaction stream.
- The supplied specification documents Kafka/Spark and alerting examples, but the available files do not establish a continuously deployed production streaming environment.
- Model metrics can change with different train/test splits, class-balancing strategies and threshold choices.

Future Improvements

- Tune classification thresholds to balance fraud detection and false-alert workload.
- Evaluate additional anomaly-detection methods such as Isolation Forest, One-Class SVM and autoencoders.
- Add automated retraining and drift monitoring so model performance can be tracked over time.
- Deploy streaming inference using a production Kafka/Spark architecture.
- Integrate secure alert delivery for high-risk transactions.
- Extend the dashboard with model-drift, threshold and investigation-history views.
- Explore future directions described in the project specification, including AI-driven credit-risk scoring, blockchain-based fraud prevention and mobile fraud alerts.

23. Conclusion

The Financial Fraud Detection project implements a complete analytical workflow using Python, Pandas, NumPy, Scikit-learn and Power BI. The five notebooks document the progression from raw CSV data through cleaning, exploratory analysis, feature engineering, supervised classification, fraud probability, risk categorisation and stored monitoring outputs. Across 5,000 supplied transactions, 482 are labelled fraudulent, producing a fraud rate of 9.64%.

Four supervised classifiers were compared using Accuracy, Precision, Recall, F1-Score and ROC-AUC. The monitoring workflow then converts fraud probability into Low Risk, Medium Risk and High Risk categories and maps those categories to Approve, Review and Block actions with Normal, REVIEW and ALERT statuses.

The Power BI dashboard presents the workflow through twelve focused pages covering transaction, merchant, payment, location, customer and time analysis, followed by fraud probability, risk, model performance, reinforcement learning, stored-data monitoring and transaction-level investigation.

The Real-Time Monitoring dashboard page is a refreshable stored-data monitoring simulation based on the saved analytical outputs; it is not a continuously running streaming system. The separate TXT specification is treated as reference material, while the implemented evidence in this report comes from the supplied notebooks, CSV outputs and dashboard PDF.

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